

Mahdi Ahmadi

Tempe, AZ | U.S. Lawful Permanent Resident (Green Card holder) | (480) 953-9954 | ahmadi@asu.edu
github.com/MahdiShAhmadi | linkedin.com/in/mahdi-ahmadi-640239119 |
scholar.google.com/citations?user=rhxKATcAAAAJ

Professional Summary

Machine learning researcher and Ph.D. student in Computer Science at Arizona State University with 4+ years of prior industry experience building production ML systems. Background spans probabilistic modeling, generative AI, computer vision, NLP, causal inference, recommendation systems, anomaly detection, and scalable ML workflows, with both academic research and shipped production work.

Education

Arizona State University, Ph.D. in Computer Science Aug 2023 – Present
GPA: 4.00/4.00. Relevant coursework: Operating Systems, Artificial Intelligence, Knowledge Representation, Probabilistic Reasoning & Learning, Data Structures & Algorithms.

Isfahan University of Technology, M.Sc. in Electrical and Electronics Engineering Sep 2016 – Apr 2019

Isfahan University of Technology, B.Sc. in Electrical and Electronics Engineering Sep 2012 – Apr 2016

Technical Skills

Programming: Python, SQL, C, C++, MATLAB

Machine Learning: PyTorch, TensorFlow, scikit-learn, Pandas, PySpark, NLP, computer vision, causal inference, recommendation systems, anomaly detection, probabilistic modeling

Systems and Tools: Linux CLI, Docker, Git, Airflow, OpenCV, FAISS, Elasticsearch, Milvus, Grafana, Kubernetes

Professional Experience

Graduate Research Associate, Arizona State University Aug 2023 – Present

- Developed probabilistic modeling methods for fair image generation, resulting in a publication at the SafeGenAI Workshop at NeurIPS 2024.
- Build and evaluate research pipelines in PyTorch and TensorFlow for computer vision and NLP applications.
- Develop tractable probabilistic-modeling methods for individual treatment-effect estimation and counterfactual inference.
- Partner on predictive modeling work with the Arizona Department of Emergency and Military Affairs for natural-disaster forecasting.

Graduate Teaching Assistant, Arizona State University Aug 2024 – May 2025

- Supported CSE 571 (Artificial Intelligence) and CSE 471 (Introduction to AI) through office hours, grading, and technical guidance.
- Helped students strengthen their understanding of search, reasoning, and practical AI problem solving.

Data Scientist, Divar 2019 – 2023

- Designed, trained, shadow-tested, and deployed an automated advertisement acceptance system across 7 verticals, reaching 25% automation and saving work equivalent to 50 operations employees.
- Developed a fraud detection model for the chat platform, improving precision by 8%.
- Built a harassment detection model using text and numerical features, improving precision from 0.26 to 0.44.
- Developed an anomaly detection model using text embeddings and time-series behavior to identify abnormal advertisements and support business decisions.
- Designed and A/B tested collaborative filtering and content-based recommendation models, increasing job-seeker engagement by 11%.
- Built workflow tooling in Python, PySpark, and Airflow and explored FAISS, Milvus, and Elasticsearch for large-scale recommendation and retrieval use cases.
- Established weekly knowledge-sharing sessions, led data science review meetings, and interviewed candidates as part of the hiring process.

Open-Source Projects and Public Work

ReDMark

github.com/MahdiShAhmadi/ReDMark

Open-source implementation of a deep-learning watermarking framework tied to published research.

Personal Website and Project Hub

mahdishahmadi.github.io

Public website that collects publications, code links, and research background.

Honors and Service

Fulton Fellowship Award, Arizona State University

2023

Reviewer, IEEE Transactions on Multimedia and IEEE Transactions on Aerospace and Electronic Systems

2022

Second Place among M.Sc. Electrical Engineering Students, Isfahan University of Technology

2018

Selected Publications

Fair Image Generation from Pre-trained Models by Probabilistic Modeling. M. Ahmadi, J. Leland, A. Chatterjee, Y. Choi. SafeGenAI Workshop at NeurIPS, 2024.

ReDMark: Framework for Residual Diffusion Watermarking by Deep Networks. M. Ahmadi, A. Norouzi, N. Karimi, S. Samavi, A. Emami. Expert Systems with Applications, 2020.

A Learning-Inspired Strategy to Design Binary Sequences With Good Correlation Properties: SISO and MIMO Radar Systems. O. Rezaei, M. Ahmadi, M.M. Naghsh, A. Aubry, M.M. Nayebi, A. De Maio. IEEE Transactions on Aerospace and Electronic Systems, 2023.

Context-aware Saliency Detection for Image Retargeting Using Convolutional Neural Networks. M. Ahmadi, N. Karimi, S. Samavi. Multimedia Tools and Applications, 2021.

Lossless Compression of Angiogram Foreground with Visual Quality Preservation of Background. M. Ahmadi, A. Emami, M. Hajabdollahi, S.M.R. Soroushmehr, N. Karimi, S. Samavi, K. Najarian. IEEE EMBC, 2018.

A Learning Approach to Design Binary Sequences with Good Correlation Properties. O. Rezaei, M. Ahmadi, M.M. Naghsh. IEEE Radar Conference, 2020.

Artistic Instance-Aware Image Filtering by Convolutional Neural Networks. M. Tehrani, M. Bagheri, M. Ahmadi, A. Noroozi, N. Karimi, S. Samavi. International Symposium on Telecommunications, 2018.

Context-Aware Saliency Map Generation Using Semantic Segmentation. M. Ahmadi, M. Hajabdollahi, N. Karimi, S. Samavi. Iranian Conference on Electrical Engineering, 2018.